WORKED SOLUTION

STEP II 1998, Question 5

Pure • Complex Triangle Inequality

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q5

The key idea

$|z_1 - z_2|$ is the distance between two points in the Argand diagram, so the inequality for $|z_1 + z_2|$ is just the statement that one side of a triangle is no longer than the other two put together. Once we have that for n terms, the final part is a matter of bounding each term by its modulus and recognising a geometric series: a small $|z|$ makes the left-hand side too small ever to reach 1.

Solution

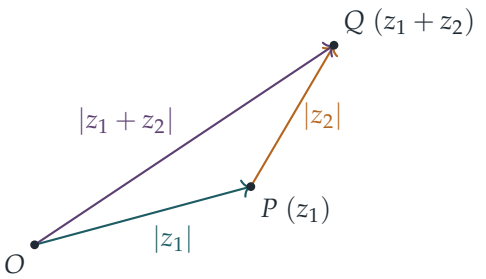
Modulus, distance and the triangle inequality

If $z = x + \mathrm{i}y$ with x, y real, the modulus of z is

$$|z| = \sqrt{x^2 + y^2},$$

the distance from the origin to the point (x, y) representing z. If $z_1 = x_1 + \mathrm{i}y_1$ and $z_2 = x_2 + \mathrm{i}y_2$, then $|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$, which is the distance between the points representing z_1 and z_2.

Now let O, P and Q be the points representing 0, z_1 and $z_1 + z_2$.



By the interpretation above,

$$OP = |z_1 - 0| = |z_1|, \qquad PQ = |(z_1 + z_2) - z_1| = |z_2|, \qquad OQ = |z_1 + z_2|.$$

The straight segment OQ is the shortest path from O to Q, so it is no longer than the path $O \to P \to Q$; that is, $OQ \le OP + PQ$, with equality exactly when P lies on the segment OQ (including the degenerate cases where the points coincide). Therefore

$$|z_1 + z_2| \le |z_1| + |z_2|.$$

The inequality for n terms

We prove by induction that for every $n \ge 1$ and all complex $z_1, \dots, z_n$,

$$|z_1 + z_2 + \cdots + z_n| \le |z_1| + |z_2| + \cdots + |z_n|.$$

For $n = 1$ both sides equal $|z_1|$. Suppose the result holds for some $n = k \ge 1$. Given $z_1, \dots, z_{k+1}$, apply the two-term inequality to the two complex numbers $w = z_1 + \cdots + z_k$ and z_{k+1}:

$$\begin{aligned} |z_1 + \cdots + z_k + z_{k+1}| = |w + z_{k+1}| &\le |w| + |z_{k+1}| \\ &\le |z_1| + \cdots + |z_k| + |z_{k+1}|, \end{aligned}$$

using the inductive hypothesis for the last step. So the result holds for $n = k + 1$, and by induction for all $n \ge 1$.

The final part

Suppose $|z| \le \frac{1}{4}$ and write $r = |z|$, so $0 \le r \le \frac{1}{4}$. Each term satisfies $|a_k z^k| = |a_k|\,|z|^k \le 3r^k$, so by the n-term inequality

$$\begin{aligned} |a_1 z + a_2 z^2 + \cdots + a_n z^n| &\le |a_1|\, r + |a_2|\, r^2 + \cdots + |a_n|\, r^n \\ &\le 3(r + r^2 + \cdots + r^n). \end{aligned}$$

> **Ask yourself** How do we turn $3(r + r^2 + \cdots + r^n)$ into something we can compare with 1, and how do we make sure the inequality is *strict*?

It is a finite geometric series, and since $r \le \frac{1}{4} < 1$ we may use the sum formula:

$$3(r + r^2 + \cdots + r^n) = \frac{3r(1 - r^n)}{1 - r}.$$

If $r = 0$ this is 0. If $0 < r \le \frac{1}{4}$, then $0 < 1 - r^n < 1$, so

$$\frac{3r(1 - r^n)}{1 - r} < \frac{3r}{1 - r} \le \frac{3 \cdot \frac{1}{4}}{1 - \frac{1}{4}} = 1,$$

where the last step holds because $\dfrac{3r}{1-r}$ is increasing in r on $[0,1)$ (its numerator increases and its positive denominator decreases). In either case

$$|a_1 z + a_2 z^2 + \cdots + a_n z^n| < 1,$$

so the left-hand side cannot equal 1. Hence the equation has no solution with $|z| \le \frac{1}{4}$.

Checks

- The bound is sharp: with every $a_k = 3$ and $z = \frac{1}{4}$ the left-hand side is $3\left(\frac{1}{4} + \cdots + \frac{1}{4^n}\right) = 1 - 4^{-n}$, for example 0.75, 0.9375, 0.984375 for $n = 1, 2, 3$. It gets as close to 1 as we like but never reaches it, which is why the strict inequality matters.
- In a random test of 200 000 choices of $n \le 15$, a_k with $|a_k| \le 3$ and z with $|z| \le \frac{1}{4}$, the largest value of $|a_1 z + \cdots + a_n z^n|$ found was about 0.996, always below 1.

Common slips

- Using the infinite sum $3r/(1-r) \le 1$ alone gives only $|\text{LHS}| \le 1$, which does not rule out a solution. The finite sum (or the fact that $r^n > 0$ when $r > 0$) supplies the strict inequality.
- In the induction, the two-term inequality must be applied to $z_1 + \cdots + z_k$ treated as a *single* complex number; that is the whole point of the step.

Beyond STEP

The triangle inequality is the defining property of a *metric*, and $|z_1 - z_2|$ makes $\mathbb{C}$ a metric space; almost every convergence argument in analysis rests on it. The final part is a baby version of results that locate the zeros of polynomials and power series: here $f(z) = a_1 z + \cdots + a_n z^n - 1$ has no zeros in the disc $|z| \le \frac{1}{4}$, in the spirit of Rouché's theorem. For the geometry of complex numbers, read Needham, *Visual Complex Analysis*; for metric ideas in analysis, Abbott, *Understanding Analysis*.

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